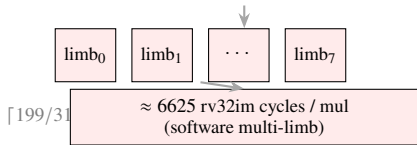


one $\mathbb{F}_p$ multiplication
 $a \cdot b \bmod p$, p a 199-bit prime

without precompile: emulate over BabyBear $p_{\text{vm}} = 2^{31} - 2^{27} + 1$



with precompile: one syscall to a native AIR

guest: **syscall**
UINT256_MUL / Griffin $_{\mathbb{F}_p^{192}}$

heavy work off the main trace

native AIR chip
(secondary trace)

≈ 1826 cycles / mul
(-70.5% vs emulated)

field-mismatch tax
 $\approx 3.6\times$ at the per-op level